Vertex N



N-type i-TOPCon

BACKSHEET MONOCRYSTALLINE MODULE

TSM-XXXNE19R **600-630W**

630W / MAXIMUM POWER OUTPUT

23.3% / MAXIMUM EFFICIENCY

High customer value

- Standardized module size with flagship module power, 30W higher compared with conventional technology
- Low Voltage design with higher string power, effectively reducing BOS (Balance of System) and LCOE (Levelized Cost of Energy) by 1%~5%
- Suitable for all scenario, especially C&I, residential, and ground applications
- Higher container space utilization effectively reduces the freight cost
- Excellent compatibility with existing mainstream system components

High power up to 630W

- Up to 23.3% module efficiency , on 210 innovation platform
- Patented i-TOPCon technology with continuous efficiency improvement, including contact resistance reduction, rear reflection enhancement and edge quality repairment

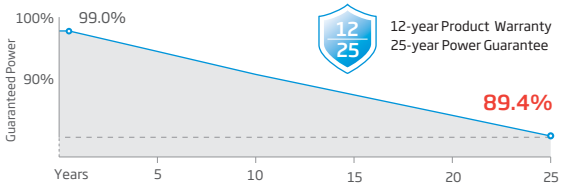
High reliability

- Minimized micro-cracks with innovative non-destructive cutting technology and high-density packaging
- Reduced risks of hot-spot with half-cut technology
- Fire Class rating C , Safety Class II

High energy yield

- Excellent low irradiation performance, validated by 3rd party
- Lower temperature coefficient (-0.29%/°C)

Performance Warranty



* Please refer to product warranty for details

Comprehensive Products and System Certificates

IEC61215/IEC61730

ISO 9001: Quality Management System

ISO 14001: Environmental Management System

ISO14064: Greenhouse Gases Emissions Verification

ISO45001: Occupational Health and Safety Management System

CAUTION: READ SAFETY AND INSTALLATION INSTRUCTIONS BEFORE USING THE PRODUCT.
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Version number: TSM_AUS_EN_2024_B
Country of Origin: China

Trinasolar

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N-type i-TOPCon BACKSHEET MONOCRYSTALLINE MODULE

ELECTRICAL DATA (STC) TSM-XXXNE19R(XXX=600-630)

Peak Power Watts-PMAX(Wp)*	600	605	610	615	620	625	630
Power Selection (W)				0 ~ +5			
Maximum Power Voltage-VMPP (V)	40.3	40.5	40.8	41.1	41.4	41.6	42.0
Maximum Power Current-IMPP (A)	14.91	14.94	14.96	14.98	14.99	15.00	15.01
Open Circuit Voltage-VOC (V)	48.4	48.7	49.0	49.3	49.6	49.8	50.3
Short Circuit Current-ISC (A)	15.80	15.83	15.86	15.89	15.91	15.93	15.94
Module Efficiency η m (%)	22.2	22.4	22.6	22.8	23.0	23.1	23.3

STC: Irradiance 1000W/m², Cell Temperature 25°C, Air Mass AM1.5. *Measuring tolerance: Pmax±3%, Voc±3% and Isc ±4%

ELECTRICAL DATA (NOCT)

Peak Power Watts-PMAX(Wp)	459	462	466	470	474	477	482
Maximum Power Voltage-VMPP (V)	37.9	38.1	38.3	38.6	38.8	39.0	39.4
Maximum Power Current-IMPP (A)	12.11	12.13	12.16	12.19	12.20	12.21	12.23
Open Circuit Voltage-VOC (V)	46.0	46.2	46.5	46.8	47.1	47.3	47.6
Short Circuit Current-ISC (A)	12.73	12.75	12.78	12.80	12.82	12.84	12.85

NOCT: Irradiance at 800W/m², Ambient Temperature 20°C, Wind Speed 1m/s.

TEMPERATURE RATINGS

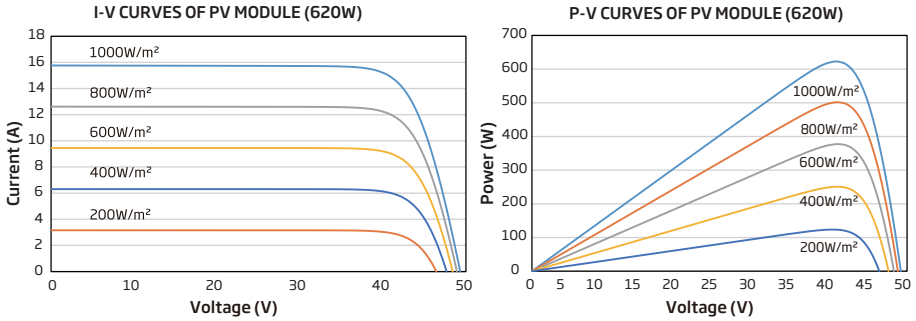
NOCT (Nominal Operating Cell Temperature)	43°C (±2°C)
Temperature Coefficient of PMAX	- 0.29% /°C
Temperature Coefficient of VOC	- 0.24% /°C
Temperature Coefficient of ISC	0.04% /°C

Due to different testing methods, the actual performances might differ from the declared specifications.

MAXIMUM RATINGS

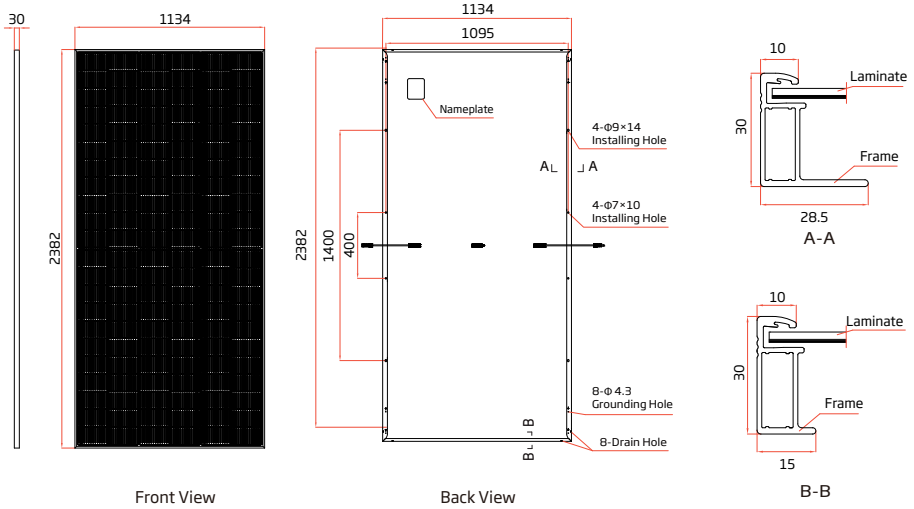
Operational Temperature	-40~+70 °C
Maximum System Voltage	1500V DC (IEC)
Max Series Fuse Rating	30A

CURVES OF PV MODULE



MECHANICAL DATA

Solar Cells	N-type i-TOPCon Monocrystalline
No. of cells	132 cells
Module Dimensions	2382×1134×30 mm (93.78×44.65×1.18 inches)
Weight	27.9 kg (61.51 lb)
Front Glass	3.2mm (0.13inches), AR Coating Tempered Glass
Backsheet	White
Frame	30mm(1.18 inches) Anodized Aluminium Alloy
J-Box	IP 68 rated
Cables	Photovoltaic Technology Cable 4.0mm² (0.006 inches²) Portrait: 350/280 mm(13.78/11.02 inches) Length can be customized
Connector	Stabuli PV-KST4-EVO2/xy_UR; PV-KBT4-EVO2/xy_UR PV-KST4-EVO2A/xy; PV-KBT4-EVO2A/xy
Packaging	Modules per box: 36 pieces Modules per 40' container: 720 pieces



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